Technische Universität Wien

188.429 – Business Intelligence (VU 4.0)

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Assignment 2 Report

ETL, Cube Modeling and Analytics (SQL/MDX/Atoti)

Group Number:	008
Student A:	Milica Aleksic 12424821
Student B:	Vidak Grujic 12332263
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1. ETL Summary

Our data warehouse uses four dimensions (Time, City, Parameter, and Alert Peak) and one fact table at the grain month × city × parameter. The dimension tables were loaded from the staging schema. Time table contains one row per calendar month. City table maps each city to its country. Parameter table stores basic information about each pollution parameter. At the end alert-peak table contains a fixed list of the five alert levels (None, Yellow, Orange, Red, Crimson). We used a simple integer key `YYYYMM` for months, because it makes joins and filtering much easier. Pre-generating the time dimension also helped us compute missing days correctly.

The fact table is built using a step-by-step SQL script. First, raw readings are combined with city, country and parameter information. Then we use alert thresholds and calculate a daily alert rank (0–4) for each reading. We then aggregate daily values to monthly level, computing exceed days, monthly peak alert level, reading counts, device counts, total data volume, averages and the 95th percentile. Finally, the results are matched to the dimension tables to replace names with surrogate keys, and missing days are calculated using the time dimension. The main difficulties were keeping the city and parameter names consistent between staging and dimensions, to also avoid double-counting when several readings occurred per day, and to deal with months where some city–parameter pairs had very few or no readings.

2. Answers to Business Questions

Bussines question selection

SQL – Student A: Q4, Q5, Q9, Q10, Q13
SQL – Student B: Q15, Q18, Q21, Q25, Q27
MDX – Student A: Q3, Q6, Q7, Q8, Q11
MDX – Student B: Q17, Q19, Q23, Q24, Q28
Atoti UI dashboards : Q1, Q2, Q12, Q16, Q22

Reflection on query implementation

- **Hierarchies vs. joins (SQL Q4, Q5 vs. MDX Q3, Q7).** SQL queries needed several joins and `GROUP BY` clauses, plus CASE statements to create quarterly or yearly columns. In MDX, the cube already contains the time and geography hierarchies, so selecting cities, countries or years only required one simple expression without joins.

- **Top–N patterns (SQL Q9, Q10 vs. MDX Q6, Q28).** To get the Top 10 results in SQL, we had to use subqueries, ORDER BY and LIMIT. Q9 even needed two WITH Statements and manual pivoting. In MDX, ranking was much shorter because of `TOPCOUNT`, which handled Top 10 logic directly and matched the wording of the business questions.

- **Filtering by alert level or category (SQL Q13 vs. MDX Q11).** In SQL, filtering by alert levels or parameter categories required extra joins and conditions. In MDX, these attributes were already part of the cube, so Q11 could simply slice the category, making the query easier to write.

- **Region filtering (SQL Q15, Q27 vs. MDX Q19).** SQL used text filters like `region_name` or manual lists of country names. In MDX, regions are part of the geography hierarchy, so Q19 could select "Central Europe" by navigating the hierarchy instead of typing country names, which made the query shorter.
- **Quarterly and monthly layouts (SQL Q9, Q13 vs. MDX Q19).** SQL needed grouping by derived date columns and sometimes CASE expressions to build quarter or month columns. MDX handled similar layouts more naturally by using the time hierarchy or a simple year slicer, so much less manual formatting was required.
- **Atoti dashboards (Q1, Q2, Q12, Q16, Q22).** Building dashboards in Atoti showed how helpful a clean star schema is. Once the cube was loaded, the creation of visual tables (e.g. region × quarter or city × alert level) required no coding. The drag-and-drop interface matched the cube's hierarchies well which made dashboard creation very quick.

Important note on Atoti dashboards:
The *Export PDF* feature in the Atoti UI does not always capture the full dashboard view. For this reason, complete dashboards for Q16 and Q22 are additionally provided as separate full-size screenshots. They are available under `pdf/a2_q16_full.png` and `pdf/a2_q22_full.png`. PDF versions of these dashboards, along with all other required dashboard exports, are also included in the submission.

3. Reflection and Lessons Learned

Student A: In this assignment I learned how important clean and consistent ETL process is. Working with the monthly fact table showed me how easily small issues in joins, dates or thresholds can affect final cube. I also got confidence to use several tools together (DBeaver, Jupyter and Atoti), and it became clear how the star schema and the cube depend on correct dimension keys. MDX was different from SQL, but work with hierarchies helped me understand OLAP concepts more clearly.

Student B: For me, the biggest learning was to see the full workflow from raw readings in the staging schema to interactive analysis in Atoti. Writing and debugging queries made me aware of the structure of the data warehouse and how important keys and clean dimensions are. I also found that dividing the SQL and MDX tasks between us worked well, because afterwards we could compare both approaches and understand the strengths of each language. Building dashboards in Atoti showed me how much easier analysis becomes once the cube is desined and we have drag-and-drop option.